

1ST YEAR B.TECH. MECHANICAL ENGINEERING
ODD SEMESTER EXAMINATION, JANUARY, 2024
ENGINEERING MECHANICS
MENUGES 01

[Full Marks: 80]

[Time: 3:00 hrs.]

Assume suitable data, if not provided

A. Multiple choice questions (Compulsory) –

[15X1 = 15]

1. The equivalent force-couple system at the origin O for the case shown in figure A1 is –

- a) $\vec{R} = -2F\hat{j}$ along $x = -r$ b) $\vec{R} = -F\hat{j}$ along $y = 3r$
 b) $\vec{R} = -F\hat{i}$ along $y = -r$ d) None of these

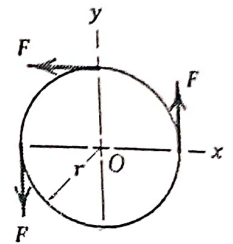


Figure A1

2. Which of the following is a vector quantity?
 a) Speed b) Energy
 c) Momentum d) None of these
3. The point, through which the whole weight of the body acts, irrespective of its position, is known as –
 a) center of mass b) centroid c) moment of inertia d) center of gravity
4. If two forces each equal to P in magnitude act at right angles, their effect may be neutralized by a third force acting along their bisector in opposite direction whose magnitude is –
 a) $2P$ b) $P/2$ c) $\sqrt{2}P$ d) $P/\sqrt{2}$
5. A framed structure with joints j is said to be perfect, if it contains members equal to –
 a) $2j - 3$ b) $3j - 2$ c) $2j - 1$ d) $j - 2$
6. Forces $5\hat{i} + 6\hat{j} + 7\hat{k}$ N and $-8\hat{i} - 9\hat{k}$ N are acting at a point. The resultant force will be of magnitude –
 a) $\sqrt{45}$ N b) $\sqrt{55}$ N c) $\sqrt{65}$ N d) $\sqrt{75}$ N
7. A body is said to be in stable equilibrium state, if the second derivative of its total potential energy is _____ zero.
 a) equal to zero b) greater than zero
 c) less than zero d) equal to one-half
8. The three forces keep acting at a point are in equilibrium, then –
 a) The forces must be coplanar b) The forces need not be coplanar
 c) The forces must be of equal magnitude d) The forces must be of unequal magnitude.
9. A mass of 10 kg is resting on a rough table. The friction force acting on it is –
 a) 0 N b) 49 N c) 98 N
 d) The question cannot be answered without knowing the co-efficient of friction.
10. The position coordinate of a particle which is confined to move along a straight line is given by $s = 2t^3 - 24t + 6$, where s is measured in meters from a convenient origin and t is in seconds. The net displacement of the particle during the interval from $t = 1$ s to $t = 4$ s is –
 a) 24 m b) 34 m c) 44 m d) 54 m
11. If the expression of linear momentum is $\vec{G} = \frac{3}{2}(t^2 + 3)\hat{j} - \frac{2}{3}(t^3 - 4)\hat{k}$, then the force will be –
 a) $t\hat{j} - 2t^2\hat{k}$ N b) $2t\hat{j} - 2t^2\hat{k}$ N c) $3t\hat{j} - 3t^2\hat{k}$ N d) $3t\hat{j} - 2t^2\hat{k}$ N

12. For a flywheel rotating freely at 60π rad/s clockwise direction while an acceleration of $4t$ rad/s² is applied counter clock wise direction. The time (t) required for the flywheel to reverse its direction of rotation is –
 a) 3.71 s b) 5.71 s c) 7.71 s d) 9.71 s
13. The angular velocity (in rad/s) of a body rotating at N revolutions per minute is –
 a) $\pi N/60$ b) $\pi N/180$ c) $2\pi N/60$ d) $2\pi N/180$
14. The total momentum of a system of masses (i.e. moving bodies) in any one direction remains constant, unless acted upon by a external force in that direction. This statement is called –
 a) Newton's first law of motion
 b) Newton's second law of motion
 c) Principle of conservation of energy
 d) Principle of conservation of momentum
15. A smooth cylinder lying on a _____ is in neutral equilibrium.
 a) concave surface b) convex surface
 c) vertical surface d) horizontal surface

B. Short questions (Answer any five questions):

[5X5 = 25]

1. Two tugboats are pulling a barge (**Figure B1**). If the resultant of the forces exerted by the tugboats is a 5000 N force directed along the axis of the barge, determine –
 a) The tension in each of the ropes, given that $\alpha = 45^\circ$
 b) The value of α for which the tension in rope 2 is minimum.

[2+3=5]

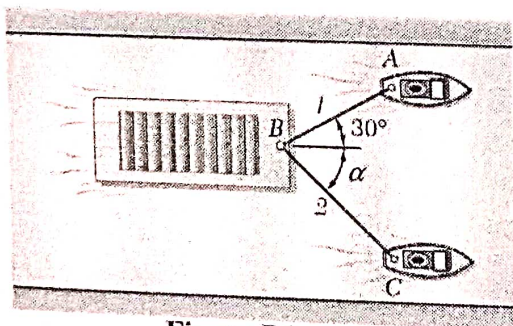


Figure B1

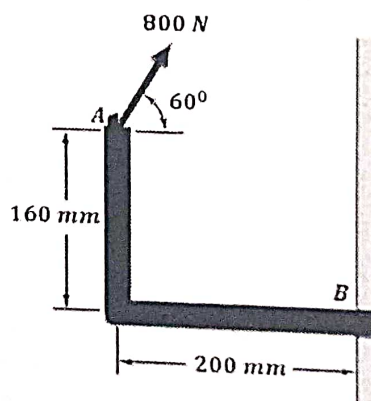


Figure B2

2. A force of 800 N acts on a bracket as shown in **figure B2**. Determine the moment of the force about B.

[5]

3. Calculate the volume V of the solid generated by revolving the 60 mm right triangular area through 180° about the z -axis (**Figure B3**). If this body were constructed of steel, what would be its mass m ?

[5]

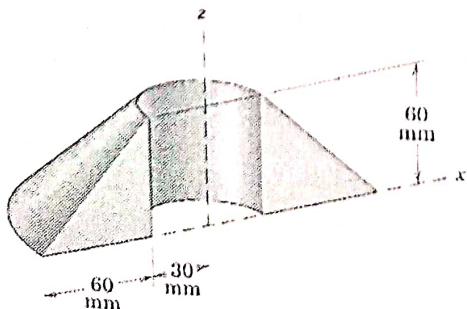


Figure B3

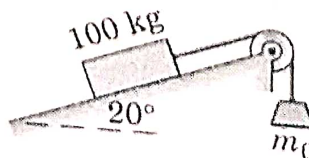


Figure B4

4. Determine the range of values which the mass m_0 may have so that the 100 kg block shown in the **Figure B4** will neither start moving up the plane nor slip down the plane. The coefficient of static friction for the contact surface is 0.30. [5]
5. A 120 kg baseball is pitched with a velocity of 24 m/s toward a batter. After the ball is hit by the bat B, it has a velocity of 36 m/s in the direction shown in **figure B5**. If the bat and ball are in contact for 0.015s, determine the average impulsive force exerted on the ball during the impact. [5]

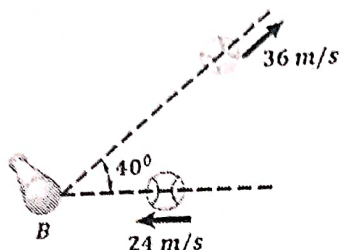


Figure B5

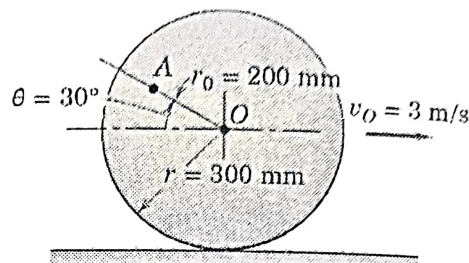


Figure B6

6. The wheel of radius $r = 300 \text{ mm}$ rolls to the right without slipping and has a velocity $v_0 = 3 \text{ m/s}$ of its center O (**Figure B6**). Calculate the velocity of point A on the wheel for the instant represented. [5]

C. Long questions (Answer any four questions):

[4X10 = 40]

1. Determine the resultant of the four forces and one couple which act on the plate shown in **Figure C1** [10]

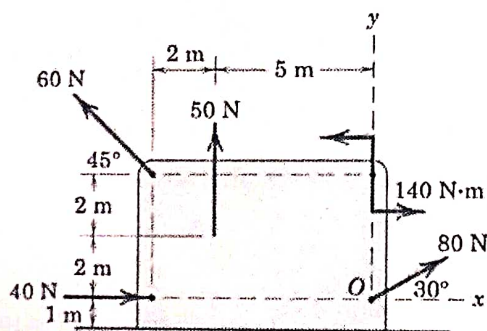


Figure C1

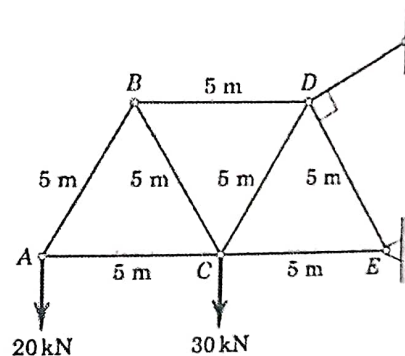


Figure C2

2. Compute the force in each member of the loaded cantilever truss by the method of joints as shown in **Figure C2**. [10]
3. The two-uniform links, each of mass m , are in the vertical plane and are connected and constrained as shown in **figure C3**. As the angle θ between the links increases with the application of the horizontal force P , the light rod, which is connected at A and passes through a pivoted collar at B , compresses the spring of stiffness k . If the uncompressed in the position where $\theta = 0$, determine the force P which will produce equilibrium at the angle θ . [10]